

Cultivating Sympoietic Literacy: Reimagining Design Education in the AI Era

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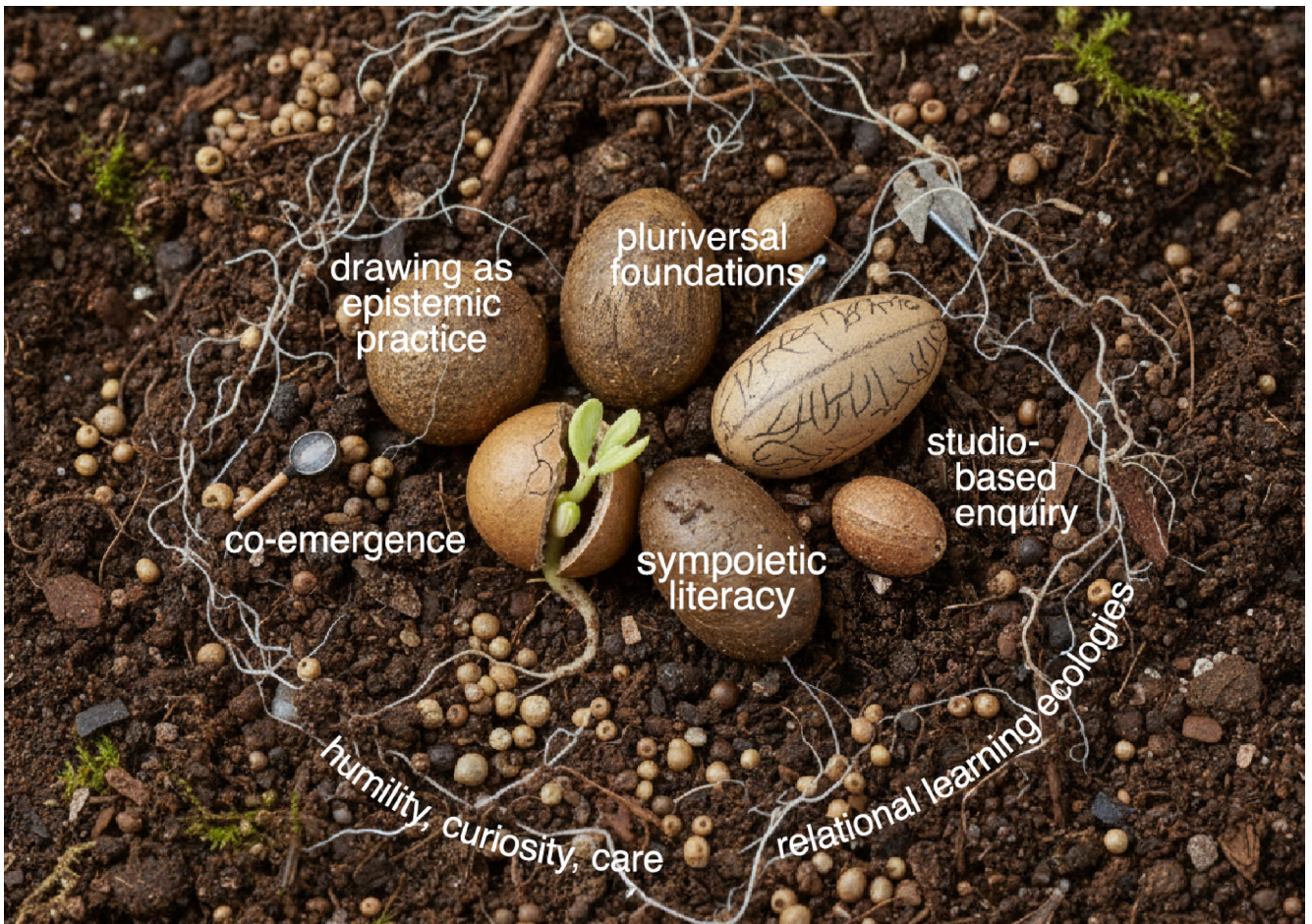
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KEYWORDS

AI-Mediated Learning, Immersive Learning Environments, NLP-Based education, Design Education, Pluriversal Pedagogy

ABSTRACT

Generative AI is reshaping design education, but current approaches treat it primarily as a productivity tool, a means to accelerate workflows, personalise content, or automate tasks. This instrumental framing inherits limitations from human-centred and Eurocentric paradigms that position creativity as individual authorship and intelligence as human property. This paper proposes an alternative: reimagining AI in design education through sympoietic literacy, a framework that treats learning-with-AI as relational practice across human, non-human, and computational intelligences. Sympoietic literacy, drawing on Haraway's concept of "making-with", shifts attention from what AI can do for learners to how humans and AI co-evolve through interaction. It involves three capacities: recognising intelligence as distributed across actors and environments; engaging uncertainty and emergence rather than seeking algorithmic control; and cultivating ethical attentiveness to how technologies encode particular worldviews. The framework is informed by trans-cultural perspectives, including the Chinese philosophical principle of *Tiān Rén Hé Yī* (human-nature unity), Yuk Hui's cosmotechnical thought, and Simondon's relational ontology, which challenge Western assumptions about technology as neutral instrument and creativity as an exclusively human endeavour. The paper reimagines the design studio as an immersive learning ecology oriented toward relational experimentation. Drawing from studio-based teaching and learning with undergraduate and postgraduate students in Sydney, it shows how practices such as drawing, narrative construction, visual iteration, and self-directed inquiry cultivate encounters with AI as a reflective collaborator, unsettling inherited assumptions of authorship, agency, and control. The study concludes that to teach and learn with AI is to participate in the broader metabolism of intelligence as a planetary process of co-learning. This calls for humility about human exceptionalism, curiosity about other ways of knowing, and care for the planetary entanglements that AI systems both depend on and shape.



Introduction

The entry of AI into design education provokes enthusiasm and unease. Discourse often frames AI in terms of automated efficiency and personalisation but conversations in design studios take different shapes. Students pose questions on authorship, integrity, and AI-generated design assets constituting plagiarism. Some see AI as creative partner. Others reject it outright.

The risk isn't that AI replaces designers, but that we train students to see design outcomes as something delivered by systems and not enacted in praxis. At stake is not only a shift in workflow, but how responsibility, values, and decision-making are understood. Design is *world-making*. If judgement is deferred to AI, students are less likely to grapple with the complex social, cultural, and ecological consequences of design choices. Design studios are critical spaces where experiential learning unfolds through engagement with imagination,

uncertainty, collaboration, materials, and co-emergence. By treating AI as a productivity tool or a policy compliance issue, we abandon what makes design education valuable.

How, when, where, why, and *if* designers use AI are critical questions requiring new reflexive literacies. The stakes extend beyond pedagogy: AI cannot be separated from ecology, labour, energy extraction, and planetary metabolism. How we frame AI will shape whether students reproduce extractive paradigms or *cultivate* capacities to participate responsibly in the socio-technical systems their design work inevitably affects.

Despite their differences, most positions on AI in design education remain grounded in human-centred, Eurocentric assumptions. Creativity is seen as a distinctly human capacity; authorship as belonging to individuals. This limits how we interpret generative systems, reducing complex interactions to matters of institutional compliance, production efficiencies, or concern about AI's impact

Sympoietic Literacy: conceptual seeds in studio ecologies. Source: Ian McArthur (2025)

on originality.

Recent work on extended intelligence (EI) offers an alternative (Pierre & Perrocheau, 2024; Barack & Jaegle, 2022; Wienrich & Latoschik, 2021). Instead of locating intelligence within discrete agents, EI posits it as emerging through relations among bodies, tools, environments, and practices. EI refers to intelligence distributed across human, non-human, and computational actors, and the cultural and material conditions connecting them. Such post-anthropocentric perspectives (Zanotti, 2024; Rijos, 2024) resonate with design education, where thinking and creativity are distributed in situated collaborative externalisation of ideas, experiences, form, image, and iterative dialogue with materials and media.

This paper introduces *sympoietic literacy* (SL) to rethink learning-with-AI in design studios. SL is informed by relational ontologies and cross-cultural perspectives, including Haraway's concept of sympoiesis or "making-with," (2015), Simondon's theory of individuation as relational emergence (2005), an expanded sense of Escobar's pluriverse (2018) as inferred in the cosmotechnical culturally situated understandings of technology proposed by Yuk Hui (2019), and the Chinese philosophy of Tiān Rén Hé Yī (human-nature unity). Together, they suggest intelligence arises through reflexive participation and not individual possession, challenging Western assumptions about technology as neutral instrument and creativity as exclusively human. Within this view, AI is not an autonomous agent nor neutral tool, but part of an extended ecology where meaning, agency, and responsibility are continually negotiated.

Recent ideas about AI literacy in education, such as that proposed by Long and Magerko (2020), offer useful taxonomies of competencies learners need to critically engage with AI. SL builds on this while shifting the emphasis from cognitive competency toward relational practice. Instead of asking what students need to *know about* AI, SL asks how students can learn to *make with* EI in ways that are ethically attentive, culturally situated, and open to co-emergence.

The design studio is a fertile context for exploring this shift (Karadağ & Ozar,

2025). Studio teaching emphasises exploratory making and embodied inquiry within the evolution of ideas and concepts through drawing, visual iteration, narrative construction, reflection and critique. As EI enters the studio, encounters with it do more than expand technical possibilities. They surface underlying assumptions about creativity and authorship, while creating opportunities to engage intelligence as something that unfolds through critically informed interaction and reflexivity. Drawing and imaging with EI can function not only as representational techniques but as epistemic practices in which students learn to work within emergent uncertainty and variation.

This work contributes to discourse on AI in education by positing SL as a speculative pedagogical modality, and by reflecting on how design educators might cultivate this orientation in their teaching. The ideas are based on observation, experiments, and reflection on studio teaching with recent cohorts of undergraduate (UG) and postgraduate (PG) design students in Sydney, Australia.

Aims and Objectives

This enquiry aims to:

- **articulate sympoietic literacy (SL)** as a design pedagogy grounded in extended intelligence (EI) and relational, trans-cultural ontologies, challenging Eurocentric and human-centred assumptions about design, authorship, ethics, and intelligence.
- **reflect on how SL tentatively emerges in studio practice**, drawing on undergraduate and postgraduate design studios at UNSW Art & Design, and on practices such as drawing, narrative construction, visual iteration, and self-directed inquiry.
- **offer preliminary insights** into cultivating attentiveness to distributed intelligence, uncertainty, emergence, and situated ethical judgement, positioning SL as an evolving orientation rather than a prescriptive model.

Conceptual Framework

The conceptual framework draws on a convergent body of thought spanning philosophies of technology, cross-cultural cosmology, and feminist technoscience. While from distinct traditions, these perspectives share a rejection of individualist and instrumental perspectives on creativity and technology, offering an understanding of intelligence and making as relational and emergent.

Simondon's relational ontology (2005) challenges assumptions that individuals - whether humans, objects, or technical systems, pre-exist their relations. He ar-

gued beings come into form through ongoing processes of individuation, shaped by their interactions with other entities and environments. Applied to technology, this unsettles ideas of technical objects as neutral tools awaiting human use. Technologies don't act independently, but participate in processes of becoming alongside humans, contributing constraints, tendencies, and potentials that shape creative outcomes through their interactions with human agents and environments.

Yuk Hui (2019) theorises the cultural and cosmological specificity of technical de-

Fig. 1 - Sympoietic Literacy: conceptual seeds in studio ecologies.
Source: Author (2025)



velopment. Hui's concept of cosmotechnics highlights how technical practices are inseparable from underlying cultural values and cosmological orders (Qiu, 2025). What is often presented as universal "technology" is, from Hui's perspective, a historically specific European cosmotechnics that has been globalised. Recognising this opens space for pluriversal technological imaginaries grounded in diverse cultural frameworks (Escobar et al., 2022).

This is reflected in the Chinese philosophical principle of *Tiān Rén Hé Yī* (天人合一), which emphasises continuity and mutual responsiveness between humans and the natural world (Shao, 2023). Rather than positioning humans as external controllers of nature, this worldview understands technical activity as participation within broader ecological and cosmic processes. Such perspectives challenge extractive and mastery-oriented models of technological progress. These Chinese philosophical perspectives are also informed by the author's long-term engagement with design education and practice in China, where theory and practice are often imbued by such themes and concerns.

Feminist technoscience complements these ideas by foregrounding situated relational practice. Haraway's sympoietic notion (2015) of "*making-with*" asserts creative and technical activities are always entangled with human and non-human others, and ethical responsibility emerges from these entanglements.

Together, these perspectives substantiate an understanding of intelligence, creativity, and agency as emerging from relations among humans, technologies, environments, and cultural contexts - not as properties located within individuals or systems.

Provisional Orientations

SL is mycelial, extending beyond the studio, toward questions of authorship, agency, responsibility, and how intelligence itself is understood in more-than-human terms. SL disrupts human-centred design by reframing intelligence as distributed across human, non-human, and technological relations. It is not just an 'ethical add-on' to design education, but responds to the fact that design operates in entangled, planetary systems that

cannot be grasped through instrumental thinking alone.

The demands of our current moment point designers toward cultivation of meta-languages: ways of observing, documenting, and articulating design intent that support their voices in dialogue with EI (Christoforou, 2025). This includes pattern recognition, probabilistic thinking, and shared vocabularies encoding values, ethical positions, and situated expertise. The question is how can design educators introduce this unfamiliar terrain to students. Conceptually, SL can be understood as provisional and developing, less a framework to be implemented than 'seeds' sown in studio ecologies (Fig. 1).

Given the rapid evolution of AI there is risk in prematurely stabilising pedagogical principles. The propositions offered are framed as an invitation: to experimentation, reflection, and ongoing negotiation among students, educators, and technologies, not a definitive account of what SL is or must become. Instead of positing a stable definition or pedagogical model, the provisional principles listed (Tab. 1) describe SL as a nascent orientation still evolving through practice.

Methods

The enquiry is grounded in studio teaching and reflective practice. SL formed gradually through curriculum development and sustained teaching and learning in UG and PG design studios where AI is introduced in discussions on research methods, place, culture, visual rhetoric, semiotics, AI, ethics and responsibility. The UG course is an advanced graphics studio where students develop visual research and design concepts for narrative-based graphic interventions in urban places. The PG course examines the socio-cultural context of design. Both include lectures and audio-visual material presenting relational perspectives on AI. The studio activities have been designed to familiarise students with integrating AI into research, drawing, narrative construction, letterform experimentation, concept development, visual iteration, and self-directed inquiry. Instead of positioning AI as an object of technical mastery, curricula focus students on

Principle	Orientation	In practice
Relational Emergence	Intelligence, creativity, and agency arise through relations among humans, non-humans, AI systems, materials, cultural contexts, and environments - not within isolated individuals or tools.	Students notice outputs shifting in response to their own decisions, model behaviour, and cultural references simultaneously.
Process Before Output	Learning is evidenced through reflective process, curatorial intent, iteration, and documentation, with outcomes understood more as provisional traces of ongoing inquiry than final resolutions.	Iteration trails, prompt logs, and curation decisions are treated as evidence of learning alongside final outcomes.
Reflexive Authorship	Designers remain responsible for judgement and nurturing meaning-making, cultivating a situated meta-linguistic creative voice in dialogue with EI rather than deferring authorship to computational systems.	Students develop a distinctive curatorial voice, making deliberate choices about what to keep, discard, or redirect — including when to set AI aside entirely.
Situated Ethical Attentiveness	Ethical engagement is embedded in practice, requiring sensitivity to social, cultural, ecological, and technological contexts as they unfold within specific design situations.	Students raise concerns about environmental cost, cultural bias, or representational norms during studio discussion and in process documentation.
Uncertainty as Generative Condition	Ambiguity, experimentation, and friction are treated as productive conditions for learning, supporting exploration without premature closure or optimisation.	Students persist with prompts that resist their intent, finding productive insight in unexpected or failed outputs and avoiding defaulting to safer approaches.
Plural and Planetary Orientation	Design practice is situated within pluriversal ways of knowing and connected to broader planetary systems, recognising local creative acts as entangled with local and global technological and ecological processes.	Students reflect on whose knowledge systems are encoded in training data and what perspectives or futures may remain unimaginable within dominant AI systems.

how outcomes emerge through curatorial modes of interaction with systems, interfaces, and datasets. More emphasis is given to exploration, dialogue, and reflection than optimisation or correctness.

Over repeated teaching cycles, patterns emerged in how students approach AI. Some engage readily with relational approaches; others resist reconciling these perspectives with established notions of creativity and authorship. SL developed in response to these varied engagements, naming emergent capacities and refining pedagogical facilitation considering critical institutional discourse, industry adoption of AI, and employment

implications. The principles of SL have not been treated as learning outcomes to be measured. They function as orienting concepts informing how studio tasks are framed and discussed. The activities described do not instantiate SL in any complete or stable sense but are examined as sites where aspects of relational engagement, attentiveness, and distributed agency is perceptible.

Results

Students' encounters with AI have reflected a range of assumptions about design process. Some approached AI as a

Table 1 - Provisional orientations informing the conceptual framework of Sympoietic Literacy

tool to be managed, others as a shortcut to be resisted, and some as a potential collaborator. By not attempting to correct these positions, the studio treated them as entry points for discussing how agency is understood within design practice. The results refer to perceptual and pedagogical shifts not measured outcomes.

As students explored AI further, outputs increasingly appeared as something emerging through interaction as well as intent. Studio provocations involving iterative prompting, editing, and curation revealed how outcomes are shaped by model behaviour, representational conventions, cultural references, and student decisions.

In the process, EI became perceptible as distributed across human, computational, and contextual elements. Bolder students experimented with relinquishing a degree of control, embracing curation and synthesising outputs with their own training images, seeing AI as a dialogic presence in the design process. This is apparent in more advanced postgraduate experimentation (Fig. 2) using multiple platforms within ‘expanded’ studio methods. Others were frustrated when results differed from expectations, returning to ‘safer’ practices. Ethical concerns, environmental impact, and cultural bias also shaped engagement, at times resulting in hesitation or disengagement. Navigating this balance between institutional protocols and encouragement of well-evidenced creative processes has informed SL as a heutagogic framework (Blaschke, 2012).

Viewed through the EI lens, these experiences reveal how cognition and creativity operate as relational processes within broader socio-technical arrangements. Students began recognising working with AI involved reflexive critical refinement and iterative curation, going beyond prompt engineering and in-painting to shape deeper participation in the creative process. While this did not resolve tensions around authorship or responsibility, it opened a dialogic space for reflective engagement. These encounters mark a shift from treating intelligence as an individual capacity toward understanding it as arising through participation and ongoing negotiation.

These dynamics are illustrated in student

process documentation submitted for assessment. One student reflected that working with AI had helped them “learn to trust my judgement where previously I tended to doubt myself”. This observation captures gradual development of reflexive authorship and strengthened capacity for situated judgement developed through sustained dialogue with it. A PG student working on an educational board game concept (Figure 2, top) documented an extensive iterative process across multiple platforms (Adobe Firefly, Gemini Imagen 3, Google ImageFX, and KREA Flux), generating and evaluating outputs for nine design components from packaging and game tokens to chance cards and game guides. Their reflection noted that AI “created amazing designs ... but required a long process through different AI models and prompts,” while also observing that AI “was not good at website design so I had to create that from scratch.” This capacity to recognise where AI collaboration was generative and where it was not reflects the situated ethical attentiveness and reflexive authorship that SL seeks to cultivate.

An UG student attempting to generate a scene of plastic debris caught in sea anemones, tried many different prompts without achieving the intended result. This encounter with the system’s resistance proved generative: sustained engagement with what the tool would and would not do sharpened their understanding of how generative outputs are shaped by training data and representational conventions.

These encounters were uneven and partial - not evidence of fully realised SL, but moments where relational engagement, reflexive authorship, and attention to distributed agency were perceptible. The following section offers a further illustration.

Drawing and Image-Making as Epistemic Practice

One UG task involved introducing constrained generative image prompts focused on ‘drawing’ letterforms. Students generated a simple image (e.g. a pink furry letter “A”) then repeated the prompt with small variations, introducing compositional references or limiting contextual detail. They then applied these processes in their own projects (Fig. 3).

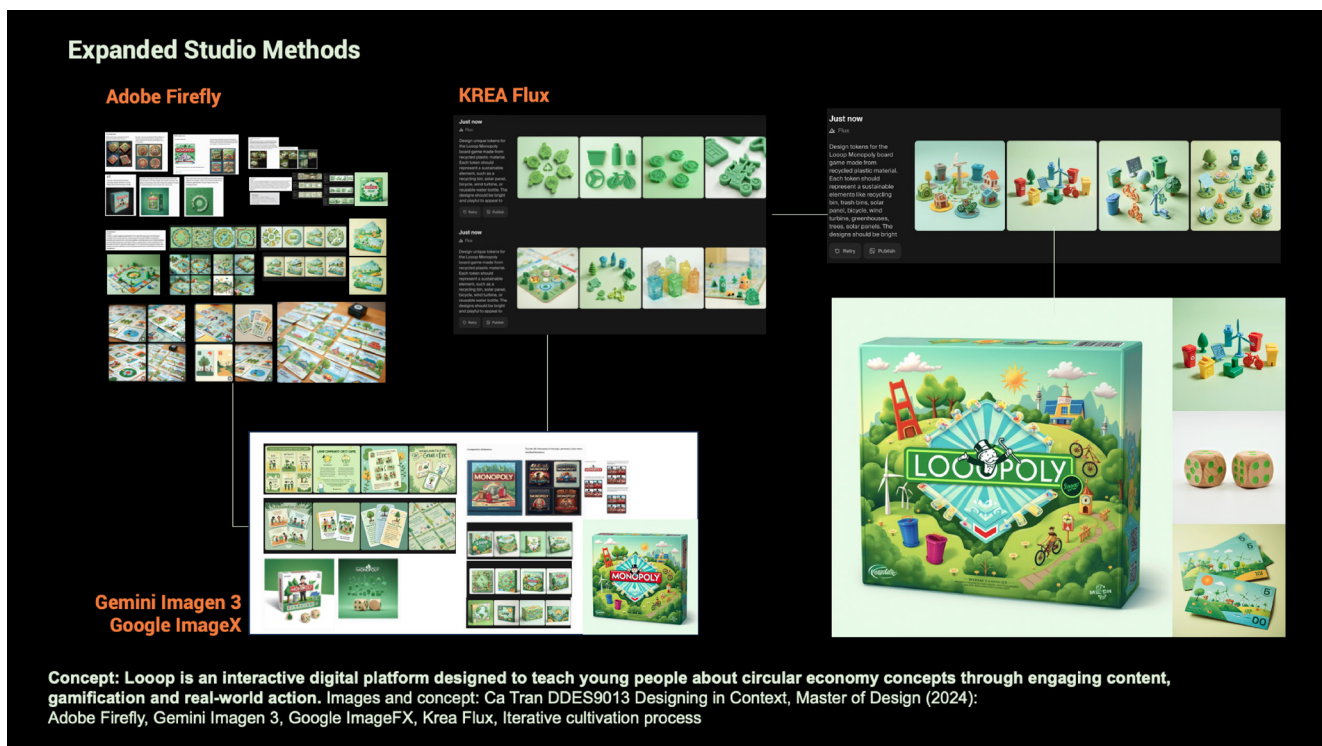


Fig. 2 - Mappings the expanded PG studio process. Source: Author (2025)

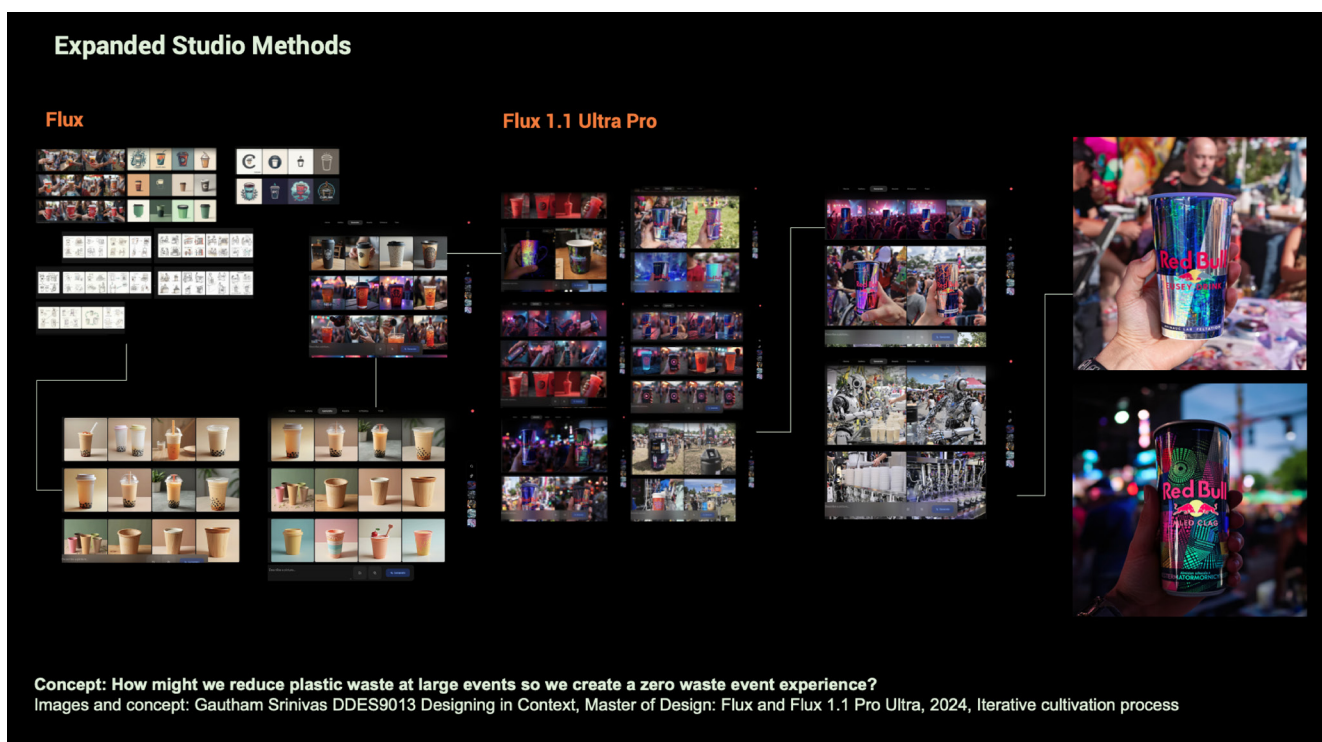




Fig. 3 - Constrained generative image iteration supporting reflective studio inquiry. Source: Author (2025)

The aim was not refined outcomes but slowing down interaction and focussing attentiveness to variation, design intent, and iterative experimentation. Students also used LLMs to explore tone of voice and alternative conceptual approaches to their place-based narratives.

They observed that small iterative prompt edits produced markedly different results. This led to discussion on how images are shaped by generative processes, training data, and representational norms not just prompt intent alone. Attention shifted from achieving desired images toward noting how differences arise. These instances of generative image-making functioned as epistemic practice comparable to drawing and thinking through making, where insight emerges through sustained engagement.

These activities make visible how EI augments studio conventions through consequential interactions between human intent and computational behaviour. Emphasising critical reflection and iterative curation over optimisation, the EI-augmented studio supports nascent SL capacities: recognising distributed intelligence, remaining with uncertainty, and reflecting on how agency is configured within contextual arrangements.

Conclusions

SL doesn't replace existing studio pedagogies. It foregrounds the relational, ecological, and reflexive conditions through which creativity and agency emerge when entangled with generative systems.

Engagement with AI did not follow a uniform trajectory. Relational approaches developed unevenly through tension, hesitation, or uncertainty suggesting studio-based pedagogies already contain many conditions needed to support this shift. Drawing, visual iteration, and narrative construction function as epistemic sites where students interrogate how outcomes are co-shaped by intent, system logics, and representational norms, opening space for ethical questions as situated concerns.

The significance lies less in the specific tools or techniques discussed than in what is at risk if design education fails to address how intelligence and judgement are framed. When generative systems are introduced as instruments for ideation or workflow optimisation, students may learn to delegate judgement rather than cultivate it. SL does not resolve this but sustains a space in which engaging with EI is something practiced, negotiated, and shared.

While the orientations described here emerged from a specific institutional context, the provisional and non-prescriptive character of SL is itself a condition of potential transferability. Educators in different contexts are invited to test, adapt, and contest these orientations not just implement them as a fixed model.

Design is not peripheral to contemporary technological and ecological crises; it is one of the mechanisms through which these conditions are produced, stabilised, and normalised. AI systems, platforms, infrastructures, and representations are all designed artefacts. Design

education therefore cannot meaningfully address AI as a ‘tool’ without also addressing the kinds of worlds that tool participates in making.

AI Usage

ChatGPT (GPT-5.2, OpenAI) and Claude (Sonnet 4.5) were used for limited editorial refinement and iterative condensation. All theoretical framing, pedagogical interpretation, and conclusions are the author’s own.

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